

EXPERIMENT – 22

Encryption and Decryption Using CBC Mode with S-DES

CODE:

```
#include <stdio.h>
```

```
void print(int a[], int n) {  
    for (int i = 0; i < n; i++) {  
        printf("%d", a[i]);  
        if (i == 3 || i == 7) printf(" ");  
    }  
}
```

```
void xorBlock(int a[], int b[], int r[]) {  
    for (int i = 0; i < 8; i++)  
        r[i] = a[i] ^ b[i];  
}
```

```
/* Simple S-DES function for demonstration */
```

```
void sdes(int in[], int out[]) {  
    int k[8] = {0,1,1,1,1,1,1,0};  
  
    for (int i = 0; i < 8; i++)  
        out[i] = in[i] ^ k[i];  
}
```

```
int main() {  
    int IV[8] = {1,0,1,0,1,0,1,0};
```

```
int P1[8] = {0,0,0,0,0,0,0,1};
```

```
int P2[8] = {0,0,1,0,0,0,1,1};
```

```
int C1[8], C2[8], temp[8];
```

```
int D1[8], D2[8];
```

```
/* CBC Encryption */
```

```
xorBlock(P1, IV, temp);
```

```
sdes(temp, C1);
```

```
xorBlock(P2, C1, temp);
```

```
sdes(temp, C2);
```

```
printf("Ciphertext: ");
```

```
print(C1, 8);
```

```
print(C2, 8);
```

```
/* CBC Decryption */
```

```
sdes(C1, temp);
```

```
xorBlock(temp, IV, D1);
```

```
sdes(C2, temp);
```

```
xorBlock(temp, C1, D2);
```

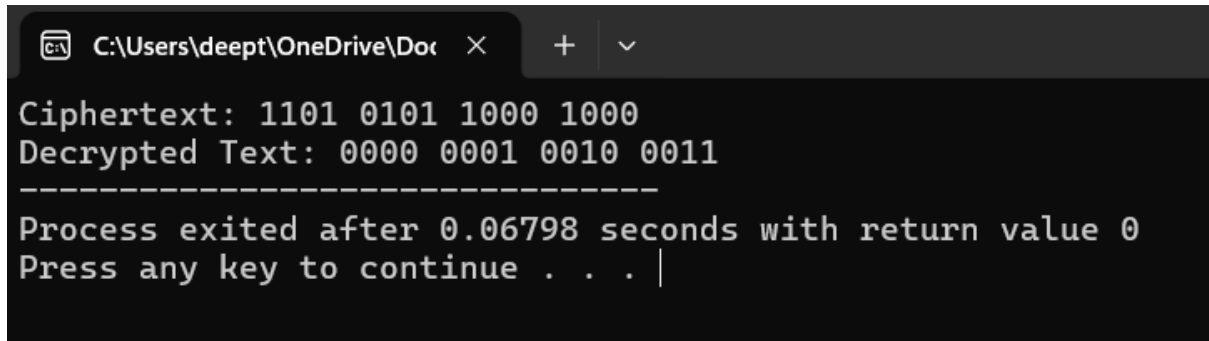
```
printf("\nDecrypted Text: ");
```

```
print(D1, 8);
```

```
print(D2, 8);
```

```
    return 0;  
}
```

OUTPUT:



```
C:\Users\deept\OneDrive\Doc >  
Ciphertext: 1101 0101 1000 1000  
Decrypted Text: 0000 0001 0010 0011  
-----  
Process exited after 0.06798 seconds with return value 0  
Press any key to continue . . . |
```